

- [CPSC 375: High-Performance Computing](#)

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Spring 2026

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Homework 25

NOTE: You are not required to hand in the following exercises, but you are strongly encouraged to complete them to strengthen your understanding of the concepts covered in class.

1. Odd-Even Transposition Sort

- A. This algorithm alternates between odd and even phases of comparisons. Explain why this sorting method is suitable for parallel execution, how many phases are needed to guarantee sorting, and what happens if one phase (odd or even) is skipped.
- B. Implement the odd-even transposition sorting algorithm using MPI to sort a distributed list of elements across multiple processes in ascending order.

2. Mergesort:

- A. Consider a parallel mergesort where recursive calls are executed in parallel. What is the parallel time complexity assuming unlimited processors, what is the total work performed, and is this algorithm work-efficient?
- B. Implement the mergesort algorithm using MPI to sort a distributed list of elements across multiple processes in ascending order, where each process stores a local portion of the array, performs local sorting, and then cooperatively merges data with other processes using message passing until the entire list is globally sorted.

3. Explain why parallel sorting algorithms require synchronization between phases, how synchronization impacts performance, and which of these algorithms is most affected by synchronization overhead.

• Welcome: Sean

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